<!-- Anonymized copy for double-anonymized review. Generated by misc/paper_b/tools/anonymize_prereg.py. -->

Anonymized copy of the registration. Sections 1 to 10 are the text registered by the git tag `prereg-heldout-2026-10-14`; sections 11 and 12 are the dated amendments appended afterwards, and the file contains no other change since the tag. Only the spans listed here were replaced; every other line is unchanged.

- SHA-256 of the source file: `7066ba81d5170428ecfe330426e1e70141a01d65567a908b37890b2d0e1b8d4c`
- SHA-256 of the file at the tag (sections 1 to 10): `5c5f05d88f752222859f3fb0eb722a49c86697ff58489853829661a2c2e7529a`
- Replaced spans:
  - `# Held-out screen: preregistration`
  - `decided by [Author A] the same day`
  - `[Author A]`
  - `([repository, anonymized for review])`
  - `[local working directory]/probe_v1/selection.csv`

<!-- end of anonymization header -->

Held-out screen: preregistration

Status: final, approved by [Author A] on 2026-10-14 and timestamped by the git tag prereg-heldout-2026-10-14 (section 10). At the time of the tag no adapter had scored a held-out item, no selection existed, and no held-out human value had been read. Nothing in this document has been computed on held-out human data. The only WVS-7 content read so far is variable labels, value labels, country identifiers and questionnaire text.

1. Question and estimator

Question. Do the sign-trained adapters predict how countries differ on WVS-7 items that played no part in their development, when the item-to-dimension assignment is fixed from the adapters alone, before any human value is read?

Units. 64 sign-profile adapters (the 2^6 factorial, `paper_b/factorial.py`) trained on bank 2, plus the base model. Countries enter only through the human criterion (the 42 GPS ∩ WVS-7 countries).

Model quantity. For adapter u and item i, the expected response code under the frozen scoring instrument (prompt variant v1, spec below), y_{ui}.

Probe (section 3) assigns each item a dimension d_i and a sign s_i.

Primary statistic. For each selected item i, the signed Spearman correlation $a_i = s_i \cdot \rho$(country weighted mean of item i, GPS z on d_i) over the 42 countries. The primary estimate is the mean of a_i over the selected items, $A = \text{mean}_i\, a_i$.

2. Pool construction (fixed)

Built by `paper_b/heldout/build_pool.py` at commit 1511995 from WVS-7 metadata only (`pyreadstat`, `metadataonly=True`).

Rule: `Qnnn` variable; not among the 35 items of `DPO_eval_WVS/question_map_wvs_edited.csv`; Q < 260 (demographics excluded); 2 to 11 contiguous non-negative codes; ordinal code order.

Text-review overrides (`paper_b/heldout/overrides.json`, reasons recorded per item): excluded Q56 (code order better/worse/same is not monotone), Q91–Q93 (knowledge quiz), Q111 (code 3 "Other answer"); 17 items confirmed ordinal by reading their labels; wording repaired for 10 items whose questionnaire text did not extract.

Result: 223 items. By response-code count: 2 codes 30, 3 codes 34, 4 codes 94,

5 codes 21, 7 codes 1, 8 codes 1, 10 codes 33, 11 codes 9.

Hashes:
  pool_skeleton.csv   c78faa7552a918e7cf840a726fb60c9b3fd1698259a31f6f18aff43bf91ea135
  overrides.json   d851264738aa6f44601c378362a8241bc633235c81ff5cf12484baa9522b09b4
  heldout_v1 spec   4d799b029102d7e7… (spec_sha256 field, 223 questions)

3. The probe (fixed)

`paper_b/heldout/probe_select.py`. For each item, OLS of y_ui on the 64 units with an intercept, six main effects (x_d = ±0.5) and all 15 two-way interactions; error df 42. t_d is the main-effect t statistic.

  top dimension d_i = argmax |t_d|, or "none" if max |t_d| < 2.0
  sign   s_i = sign(b_{d_i})
  imprint set   trust, risktaking, posrecip, altruism (the four surfaces with a
                 cohort-1 imprint; patience and negrecip were flat)
  PRIMARY rule   among items with d_i in the imprint set and |t| ≥ 2, the top 25%
                 of all 223 items by |t| (at most 56 items)
  SECONDARY rule d_i in the imprint set and |t| ≥ 2 (fixed threshold)

Known property, stated in advance: under pure noise, about 26% of items cross |t| ≥ 2 on at least one of six dimensions (documented by `test_null_items_are_flagged_at_the_familywise_rate`). The selection rule does not control this rate; the permutation null in section 4 does.

Decided 2026-10-14 (D5): the secondary rule is a reliability check. The primary rule alone carries the confirmatory test at one-sided alpha 0.05; the secondary rule's result is reported beside it and does not enter the inference.

4. Human criterion and inference

Human values. WVS-7 country means with the survey weight (W_WEIGHT), codes < 0 set missing, for the selected items only. Read only after FREEZE.json exists.

Inference for A.
  Permutation null over items: draw as many items as were selected, uniformly from the 223-item pool, and give them the (d, s) pairs of the selected items in random order; 10,000 draws; one-sided p = share of null A ≥ observed A. This null absorbs the familywise selection rate in section 3.
  Country bootstrap: 2,000 resamples of the 42 countries for a 95% interval on A.

Also reported (descriptive): the share of selected items with a_i > 0, and A per dimension.

5. Blinding

`paper_b/heldout/freeze.py`. `freeze()` writes FREEZE.json with the SHA-256 of selection.csv and pool_skeleton.csv, the code revision and the scoring run ids, and refuses to run twice. Every reader of held-out human data calls `require_freeze()`, which fails if FREEZE.json is absent or either frozen file has changed. Order of work: score held-out items (model only) → probe selection → freeze → human join.

6. Comparators

Random allocation null: section 4.

Face-valid rule (decided 2026-10-14). What a careful reader would assign from the questionnaire alone, built by `paper_b/heldout/comparator.py` with the fixed procedure in its docstring: a construct lexicon assigns an item to a dimension only when exactly one dimension's words occur in its wording; the direction is

read from the response labels; a person may only confirm, flip, or abstain, with a written reason per change (`comparator_review.json`). No model output and no human response were used.

Result on the 223-item pool: 30 items assigned (trust 20, risktaking 4, negrecip 4, altruism 2); 9 lexicon matches set to "none" and 3 directions flipped, each with its reason. Of these, 26 fall in the imprint set used by the primary rule.

Test. A_face is the section-1 statistic computed on the comparator's own assigned items and signs, restricted to the imprint set (26 items) as the primary rule is. The quantity of interest is D = A_adapter − A_face, with a 95% interval from the same 2,000 country-bootstrap resamples used for A (both statistics recomputed on each resample). Readings:

  D's interval above 0 → the adapters find structure the wording does not reveal
  D's interval covers 0 → the adapters match a careful reader
  D's interval below 0 → the reader's rule screens better

A_face on all 30 items (including negrecip) is reported beside it.

Known limitation, stated now: the face-valid rule is concentrated on trust (20 of 30 items, mostly the institutional-confidence battery), so D partly compares dimension coverage as well as assignment quality; per-dimension A for both routes is reported.

System One comparator (decided 2026-10-14, before any held-out item was scored by an adapter). A typed decision model that returns calibrated probabilities instead of text (TypeSafe AI, jev-1.13.0, pinned; typesafe-sdk 0.7.5) reads only an item's wording and response options, codes in ascending order (`paper_b/heldout/jev_comparator.py`, whose docstring states the rule). Call 1 is a choice over the six GPS construct definitions plus "none"; these are the definitions used to generate the training data (measurement-method clauses removed) and contain no country, model-output or human-response information, so the contrast is not circular. One run assigns an item when the top choice is not "none" and its probability is at least 0.5 (untuned). Call 2, for an assigned item only, asks whether the lowest code reveals more of that trait than the highest; sign is −1 when that probability is at least 0.5, else +1. Exact ties: probability 0.5 resolves to −1, call-1 ties resolve by definition order.

Repeat-call rule. A single call proved unstable at the threshold: two complete runs disagreed on 5 of 223 pool items, all with top probability within 0.03 of 0.5 or a direction probability within 0.04 of 0.5, while the 23 battery items were identical. The comparator of record is therefore five independent runs, and an item is assigned only when at least three runs give the same dimension and sign. All five runs' requests and answers are archived (`jev/run1`…`jev/run5`) and frozen with the selection.

Results. Battery validation (reported, never used for tuning): 12 of 23 frozen dimensions recovered (trust 9 of 9, altruism 2 of 3, negative reciprocity Q195; Q81 positive reciprocity read as trust), 12 of 12 directions correct; the other 10 items are abstentions, mostly items the earlier direction audit classed as non-aligned or ambiguous. This design cannot say whether those abstentions reflect the comparator or the battery, and neither reading is claimed. Pool: 42 of 223 items assigned (trust 33, altruism 7, risk-taking 1, negative reciprocity 1); 39 unanimous; Q210 holds its sign by 3 of 5 runs and is flagged. On the 23 items both comparators assign, they agree on dimension and sign 23 of 23.

Test. Abstentions never enter a statistic (`screen_test` refuses sign-0 rows). The primary reading of the contrast with each comparator is coverage-matched: D_common = A_adapter − A_comparator computed on the items both the adapter selection and the comparator assign, so the same items enter both statistics, with a 95% interval from the paired 2,000-resample country bootstrap. Reported beside it: D over each route's own items, D per dimension, and each route's coverage by dimension. Readings as for the face-valid rule.

Known limitations, stated now: both comparators concentrate on trust (Jev 33 of 42, face-valid 20 of 30), which is why the coverage-matched and per-dimension contrasts are primary; jev-1.13.0 is a closed model with stochastic probabilities, so the archived runs, not a re-query, are the record.

7. Human ceiling

For items fielded in both waves, the Spearman correlation of country means between WVS wave 6 and wave 7 (WVS time series, 28 countries with both waves), averaged over the selected items, with the same signed construction against GPS z. This gives the agreement two human surveys achieve on these items, the reference against which A is read.

8. Planned sensitivity

  secondary threshold rule (section 3)
  dropping items whose |a_i| < 0.15 is NOT used for the primary; reported only as item-level descriptives
  leave-one-country-out range of A
  readout variants: the probe re-estimated under reverse_options and both paraphrases; the share of items whose (d_i, s_i) is unchanged
  sign-prompt arm: the same probe applied to the 64 sign-prompt profiles

9. What would count against the claims

  Primary: A not above the permutation null at one-sided 0.05 → the adapters do not screen held-out items better than random assignment.
  A above the null but below one-third of the wave-6/7 ceiling → screening signal exists but is weak; reported as such.
  Readout variants changing (d_i, s_i) for more than half of the selected items → the selection depends on prompt wording.

10. Version

Code: the commit carrying the tag prereg-heldout-2026-10-14 on feat/paper-b-cohort ([repository, anonymized for review]). The pool was built at commit 1511995.

Timestamp (D4, decided 2026-10-14): this document in the project's GitHub repository, committed and pushed with the tag prereg-heldout-2026-10-14 before any held-out item is scored. The pushed tag and its commit date are the public record; an OSF copy may follow later.

Frozen inputs at the tag (SHA-256):

pool_skeleton.csv	c78faa7552a918e7cf840a726fb60c9b3fd1698259a31f6f18aff43bf91ea135
overrides.json	d851264738aa6f44601c378362a8241bc633235c81ff5cf12484baa9522b09b4
face_valid.csv	047c05a3c819385740bca5bf17442e95b928067be7336cdf2f9193ff3963122b
comparator_review.json	5ad6decd53cadf2a8551bff4edd3acbf9a49dc1bc323114b59a7f656c1dd7bb7
jev/majority_face_valid_jev.csv	53d0c8b8062f27c311a5bec32d133e56de877365a65ba32d3f667b2af21092c2

Approvals: the 12 face-valid review calls (9 abstentions, 3 direction flips) were approved by [Author A] on 2026-10-14; D5 (secondary rule is a reliability check) and D4 (timestamp vehicle) decided by [Author A] the same day.

11. Amendment 1 (2026-10-16): standalone wording repaired before the freeze

What happened. The first held-out scoring pass completed on the pool as registered at tag `prereg-heldout-2026-10-14` (runs `bank2-heldout-v1-20261016`, 65 files; `bank2-strict-v1-20261016`, 65 files). Reading the item text afterwards found 12 items whose standalone wording cannot be answered as written: a sentence cut off mid-clause (Q46, Q175, Q253, Q199), a question left with a stripped country placeholder (Q121, Q254), a grid row truncated to a bare pronoun or

article (Q82, Q255, Q256), and three second-choice follow-ups that refer to a question that does not exist in a standalone prompt (Q153, Q155, Q157). The same defect classes appear among the ten wording repairs already recorded in section 2 (for example Q257 and Q258, whose grid rows were truncated in exactly the same way); the pool rule and the override mechanism are unchanged.

Full disclosure. The first probe pass was computed and read BEFORE the repairs were chosen, and two of the repaired items (Q153, Q157) fall in that pass's top 25% by |t|. Nothing is frozen: `FREEZE.json` does not exist and no held-out human value has been read. The first pass output is retained at `[local working directory]/probe_v1/selection.csv` and is superseded.

What changed. 12 items (Q46, Q82, Q121, Q153, Q155, Q157, Q175, Q199, Q253, Q254, Q255, Q256), each repaired from the item's own read-out options or from its variable label, with the reason recorded per item in `overrides.json`. Pool size is unchanged at 223. Section 2's rule, the probe (section 3), the criterion (section 4) and the blinding guard (section 5) are unchanged.

Amended hashes

pool_skeleton.csv	19241f6f103c51d64e295045162bb370920223b7ffad7c5f3c8991f5c101bd44
overrides.json	0f5cd2bbd88ef6eaca9cca9ded24e8a5e056276be055b21c1d92b941dd7d3849
heldout_v1 spec_sha256	46447de63f7cc161c016bdc38a3a7c9e8d351be308c44cf1ccd28d9091d13b38
heldout_reverse spec_sha256	d30bc4aa67d15430e52e09fdd2f8dd621a4b6aeee1559b6e26abf61a22d87e6c
heldout_para_a spec_sha256	cbd63283b4d99f6b41a6119148821a56093acb2296845f0662c4185c2f115f47
heldout_para_b spec_sha256	0680dffbf3d5c67bdf040a93d8454fc3de1ec08e04afdd8fd49360a6d8b30934
signprompt (uses heldout_v1)	46447de63f7cc161c016bdc38a3a7c9e8d351be308c44cf1ccd28d9091d13b38

Superseded. Section 10's `pool_skeleton.csv` hash `c78faa75…` is the pre-amendment file (CRLF working copy; its LF git blob is `b221d12a…`, the discrepancy the verifier raised as F7). Section 10's `heldout_v1 spec 4d799b029102d7e7…` is the pre-amendment spec.

Consequences.

- The held-out battery is re-scored for all 64 adapters under the amended spec as a single run id, so one instrument covers every item. The sign-prompt arm and the three readout variants are re-scored under the amended spec too. The strict battery (23 frozen items) is untouched, because its question map is not the held-out pool; its 65 files stand.
- Open item, to be decided before the human join: the face-valid rule assignments and the System One (Jev) comparator assignments for these 12 items were made on the pre-amendment wording, since both comparator files cover all 223 items. Either they are re-derived on the amended wording, or the pre-amendment wording is disclosed for those items.

12. Amendment 2 (2026-10-17): the 12 repaired items leave the comparator routes

Decision. [Author A] chose exclude-and-disclose for the open item in section 11, in the same instruction that authorised the freeze ("exclude-and-disclose and freeze"), on 2026-10-17. The freeze was committed at 15:40 (`5982544`) and the first read of held-out human data followed at 15:43 (`83cd427`). At the time of the decision no human held-out value had been read and no comparator statistic had been computed. This text was written after the primary statistic A had been read; it records the earlier decision and changes nothing else.

Rule. The 12 items repaired by Amendment 1 (Q46, Q82, Q121, Q153, Q155, Q157, Q175, Q199, Q253, Q254, Q255, Q256) are removed from both comparator files before any comparator statistic is computed, because both comparators assigned them on the pre-amendment wording. `analyze_heldout.REPAIRED` holds the list, and a test fails if it drifts from section 11.

Effect. Of the 12, only Q82 was assigned by either comparator: trust, sign −1, by both the face-valid rule and the Jev majority. The face-valid route goes from 30 to 29 assigned items (imprint set 26 to 25) and the Jev route from 42 to 41 (imprint set 41 to 40). Q82 is not in the adapter's primary selection. The adapter route is unaffected: its scores were produced on the repaired wording (Q253, a repaired item, is among the 56 selected and stays). The frozen comparator files are unchanged on disk; the exclusion happens in code.